A380X ATC mailbox with the SD on an ND display unit

When the SD is shown on an ND display unit (DU RECONF pushbutton, e.g. with the SD DU off), the ATC mailbox comes with it and works there, as the A380 FCOM asks; the ATC MSG lights flash, the RING waits for the reading time and the KCCU MAILBOX key brings the cursor to the mailbox

Aircraft	A380X
Where to find it	F/O (or CAPT) DU RECONF pushbutton on the glareshield; the ATC mailbox at the bottom of the SD, on the SD DU or on an ND DU; ECAM CDS SD DU FAULT procedure
Branch	`core/a380x-mailbox-du-reconf`, based on `core/display-unit-failures` with `core/a380x-atccom-datalink` merged in (stacked: needs both, and their bases, merged first). `core/display-unit-failures` brings the DU reconfiguration (the SD on an ND DU) and the mouse fix for DUs holding hidden displays; `core/a380x-atccom-datalink` brings the ATC mailbox. Merge both into this branch again when they change. There is no cockpit printer here (`core/printer`): PRINT in the mailbox says PRINT NOT AVAIL.
Status	Tested in the sim by the user on a combined test build (with the mouse fix of `core/display-unit-failures`): mailbox on the F/O ND with the SD off: OK. Not sim-tested on this core branch itself. Unit tests: AtcMailbox.spec.tsx, MailboxLogic.spec.ts, MailboxBus.spec.ts, the CdsReconfiguration specs and HostedDisplay.spec.ts. ATC MSG flashing, RING delay and MAILBOX key: not sim-tested, unit tests only.
Pictures	No pictures: the mailbox on the ND DU looks like the mailbox at the bottom of the SD.

1. ATC mailbox follows the SD to the ND display unit

> **Aircraft:** A380X | **Where:** F/O DU RECONF pb; ATC mailbox on the F/O (or CAPT) ND DU | **Status:** Tested in the sim by the user on a combined test build: OK (SD off, F/O DU RECONF twice, mailbox on the F/O ND with its buttons, back on the SD with the same queue). Not sim-tested on this core branch. Unit tests: AtcMailbox.spec.tsx, MailboxLogic.spec.ts, MailboxBus.spec.ts, CdsReconfiguration specs, HostedDisplay.spec.ts.

What it is for

The ATC mailbox is the bottom section of the SD. The CDS SD DU FAULT procedure says "If the SD DU is turned OFF, use the F/O DU RECONF P/B for ATC mailbox" (A380 FCOM PRO-ABN-ECAM-10-31). On `core/display-unit-failures` the F/O DU RECONF pb brings the SD to the F/O ND DU, but without the mailbox, so the ATC messages could not be read or answered with the SD DU off. Now the mailbox comes with the SD.

What you should see

Situation	What you should see
Normal operation (SD DU on)	The mailbox at the bottom of the SD DU, as before.
SD DU off or failed, F/O DU RECONF pressed until the SD shows on the F/O ND DU	The SD with the mailbox on the F/O ND DU: the same queue and the same message; its buttons and the page arrows work with the F/O cursor.
SD DU on again	The mailbox is active on the SD DU again, with the same queue.
The SD shown both on the SD DU and on an ND DU (DU RECONF used because of another failed DU)	The SD DU's mailbox is the active one; the mailbox on the ND DU shows the same message and takes no input (design choice).

The SD shown nowhere	No mailbox marks a new message as displayed or read: the ATC MSG lights stay on until a DU shows the mailbox again (design choice).
PRINT in the mailbox	PRINT NOT AVAIL (no cockpit printer on this branch; printing is on `core/printer`).

How it works

- The systems host decides which DU has the active mailbox (`L:A380X_CDS_MAILBOX_DU`): the SD DU while it is working and shows the SD, else the ND DU that shows the SD (the SD is never on both sides at once, FCOM DSC-31-15-20), else none.
- Only the active mailbox answers the crew and sends to the ATC function; another mailbox on screen mirrors the displayed message.
- The A380X README and the simvar documentation describe the variable and the rule.

Limitations and why

Limitation	Why	What to do
Needs an ATC connection with a message to see anything in the mailbox.	The mailbox shows the messages of the FBW ATSU (`core/a380x-atccom-datalink`).	Log on to an ATC center (Hoppie or BeyondATC) or prepare a request with XFR TO MAILBOX.
No printing.	The cockpit printer is `core/printer`, not merged here.	Merge it for PRINT.
A full MSFS restart is needed after an update.	It comes with the panel.cfg and cockpit changes of `core/display-unit-failures`.	Restart the sim after installing.

How to use

1. With the SD DU off or failed (CDS SD DU FAULT), press the **F/O DU RECONF** pushbutton (glareshield, F/O side) until the F/O ND DU shows the SD: first the E/WD, then the SD with the mailbox.
2. Read and answer the ATC messages in the mailbox on the F/O ND with the F/O cursor, as on the SD.
3. Press F/O DU RECONF once more to get the ND back, or switch the SD DU back on.

How to test

#	Do this	Expected result
1	A380X powered, logged on to an ATC center (MFD ATC COM > CONNECT) with at least one message in the mailbox (e.g. a request prepared with REQUEST, XFR TO MAILBOX).	The message in the mailbox at the bottom of the SD.
2	Turn the SD DU off (ECAM control panel on the pedestal: SD knob fully left, OFF), or flyPad > Failures > ATA 31: **SD display unit**.	The SD DU goes blank (with the failure: CDS SD DU FAULT with IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX).
3	Press the F/O **DU RECONF** pushbutton twice.	The F/O ND DU shows the E/WD, then the SD with the mailbox: the same queue and the same message.
4	Move the F/O cursor to the mailbox and click the previous-message arrow (or a message button).	The mailbox reacts on the F/O ND.
5	Press **PRINT** in the mailbox.	PRINT NOT AVAIL.
6	Turn the SD DU back on (or remove the failure) and wait for its self-test.	The SD mailbox shows the same queue and its buttons work.

2. ATC MSG lights flashing, RING delay and the KCCU MAILBOX key

> **Aircraft:** A380X | **Where:** Glareshield ATC MSG pushbuttons; ATC RING aural; KCCU MAILBOX key; SD mailbox | **Status:** Not sim-tested (added; unit tests: MailboxBus, AtcMsgLights and AtcMailbox specs, failing first).

What it is for

Three items of the A380 FCOM ATC datalink description (DSC-46-10-20):

- **ATC MSG lights flash at 2 Hz** (250 ms on, 250 ms off) while an ATC message waits in the mailbox, until either ATC MSG pushbutton is pressed; then both go off (FCOM: "The pushbutton flashes at a frequency of 2 Hz"). A new variable `L:A380X_GLARESHIELD_ATC_MSG_LIGHT` drives the two pushbuttons.
- **RING delay:** the ATC RING sounds only if the message is not read within 15 s (5 s for an urgent message), then repeats. The A32NX DCDU is unchanged.
- **KCCU MAILBOX key:** puts the KCCU cursor of that side on the SD mailbox (FCOM: "When pressed, the cursor moves to the mailbox section of the SD"); the MFD of that side hides its cursor. The key works with the KCCU keyboard ON. Design choice: the cursor goes to the middle of the mailbox (the FCOM gives no position).

Situation	What you should see
An ATC uplink arrives	Both ATC MSG pushbuttons flash at 2 Hz.
Message read within 15 s (5 s if urgent)	No RING.
Message not read after 15 s (5 s if urgent)	RING, then repeated until the message is read.
Either ATC MSG pushbutton pressed	Both lights go off.
KCCU MAILBOX key pressed (keyboard ON)	The cursor of that side jumps to the SD mailbox.

Limitations and why

Limitation	Why	What to do
The light changes need a flight reload.	The glareshield behaviour XML changes.	Reload the flight after installing.
The MAILBOX key does nothing with the KCCU keyboard OFF.	FCOM: the KCCU keys work with the keyboard switch ON.	Set the KBD switch ON.

How to use

1. When the ATC MSG lights flash, press one of them (or read the message) to stop them.
2. Press the KCCU **MAILBOX** key to bring the cursor to the SD mailbox, then answer the message.

How to test

#	Do this	Expected result
1	A380X, datalink logged on (BeyondATC or Hoppie). Send yourself a normal CPDLC uplink.	Both ATC MSG pushbuttons flash about twice a second; no RING in the first 15 s.
2	Do not read it for 15 s.	RING; it repeats until the message is read.
3	Send an urgent uplink.	RING after 5 s if not read.
4	Press one ATC MSG pushbutton.	Both lights go off.
5	KCCU keyboard ON, cursor on the MFD. Press the KCCU **MAILBOX** key.	The cursor moves to the middle of the SD mailbox; the MFD cursor goes.

What PR testers should check

- ATC MSG flashing and the RING delay with BeyondATC.
- The MAILBOX key cursor (both sides, also with the SD on an ND DU).
- Not sim-tested yet.